Chapter 3 — Circular and Rotational Motion

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. One radian is approximately:

(a) 57.3° **(b)** 180° **(c)** 90° **(d)** 45°

2. One complete revolution equals:

(a) 4π rad **(b)** π/2 rad **(c)** π rad **(d)** 2π rad

3. Angular displacement is measured in:

(a) newton **(b)** radian **(c)** metre **(d)** joule

4. The relation between linear and angular speed is:

(a) $v = r^2\omega$ **(b)** $v = \omega/r$ **(c)** $v = r\omega$ **(d)** $v = r/\omega$

5. The relation between tangential acceleration and angular acceleration is:

(a) $a_t = \alpha/r$ **(b)** $a_t = r^2\alpha$ **(c)** $a_t = r/\alpha$ **(d)** $a_t = r\alpha$

6. Centripetal acceleration is:

(a) v^2/r **(b)** v/r **(c)** vr **(d)** v^2r

7. Centripetal force is:

(a) mv^2r **(b)** mv/r **(c)** mv^2/r **(d)** mvr

8. Centripetal force is directed:

(a) along tangent **(b)** opposite motion **(c)** toward the centre **(d)** away from centre

9. In uniform circular motion, speed is:

(a) increasing **(b)** zero **(c)** constant **(d)** decreasing

10. In uniform circular motion, velocity is:

(a) changing **(b)** zero **(c)** constant **(d)** maximum

11. Work done by centripetal force in uniform circular motion is:

(a) maximum **(b)** zero **(c)** mgh **(d)** mv^2/r

12. For a satellite, centripetal force is provided by:

(a) gravity **(b)** normal force **(c)** tension **(d)** friction

13. For a car turning on a level road, centripetal force is provided by:

(a) tension **(b)** air pressure **(c)** friction **(d)** gravity

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14. Orbital speed of a satellite is:

(a) √(gr) **(b)** √(GM/r) **(c)** √(2gr) **(d)** GM/r

15. Orbital speed of a satellite depends on:

(a) shape of satellite **(b)** colour of satellite **(c)** mass of Earth and radius **(d)** mass of satellite

16. Near Earth's surface, orbital speed is about:

(a) 11.2 km s^{-1} **(b)** 1.0 km s^{-1} **(c)** 7.9 km s^{-1} **(d)** 3.0 km s^{-1}

17. Near Earth's surface, orbital period is about:

(a) 60 min **(b)** 12 h **(c)** 84 min **(d)** 24 h

18. Astronauts in orbit feel weightless because:

(a) mass is zero **(b)** air is absent **(c)** they are in free fall **(d)** gravity is zero

19. Artificial gravity in a rotating station is due to:

(a) gravity of Earth **(b)** magnetic force **(c)** friction **(d)** normal force

20. Moment of inertia depends on:

(a) mass only **(b)** volume only **(c)** angular momentum only **(d)** mass distribution about axis

21. The rotational analogue of mass is:

(a) angular velocity **(b)** torque **(c)** angular momentum **(d)** moment of inertia

22. Torque equals:

(a) rF sin θ **(b)** rF cos θ **(c)** F/r **(d)** Fr^2

23. The rotational form of Newton's second law is:

(a) τ = Iω **(b)** τ = L/t **(c)** τ = Iα **(d)** τ = mr

24. Angular momentum of a rigid body is:

(a) Iω **(b)** mv **(c)** Iα **(d)** τt

25. Angular momentum is conserved when:

(a) external torque is zero **(b)** mass is constant **(c)** speed is constant **(d)** external force is zero

26. When a skater pulls arms in, angular velocity:

(a) remains same **(b)** increases **(c)** decreases **(d)** becomes zero

27. When a skater pulls arms in, moment of inertia:

(a) remains same **(b)** decreases **(c)** increases **(d)** doubles

28. A gyroscope maintains direction due to:

(a) angular momentum **(b)** linear momentum **(c)** friction **(d)** gravity

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29. A flywheel stores energy as:

(a) rotational kinetic energy **(b)** heat **(c)** potential energy **(d)** sound

30. A centrifuge separates materials using:

(a) gravity only **(b)** magnetic force **(c)** centripetal force **(d)** electric force

31. If angular velocity is constant for all particles of a rigid body, linear velocity is:

(a) different at different radii **(b)** zero **(c)** same everywhere **(d)** maximum at axis

32. The average velocity for one complete revolution is:

(a) zero **(b)** v **(c)** $2\pi r/T$ **(d)** ωr

33. The average acceleration for one complete revolution is:

(a) $\omega^2 r$ **(b)** v^2/r **(c)** zero **(d)** g

34. A stone whirled in a vertical circle breaks the string at the bottom because tension is:

(a) maximum **(b)** zero **(c)** minimum **(d)** equal to weight

35. Bubbles in a swung soda bottle collect near the:

(a) neck **(b)** bottom **(c)** middle **(d)** outside

36. The time period of a simple pendulum in an artificial satellite is:

(a) 1 s **(b)** infinite **(c)** zero **(d)** $2\pi\sqrt{(l/g)}$

37. The SI unit of angular acceleration is:

(a) rad s^{-1} **(b)** m s^{-2} **(c)** s^{-1} **(d)** rad s^{-2}

38. The SI unit of torque is:

(a) kg m s^{-1} **(b)** N s **(c)** N m **(d)** J s

39. The SI unit of angular momentum is:

(a) kg m^2 s^{-1} **(b)** kg m s^{-1} **(c)** N m s **(d)** J s^{-1}

40. The dimensions of moment of inertia are:

(a) $[ML^2T^{-2}]$ **(b)** $[ML^2]$ **(c)** $[MLT^{-2}]$ **(d)** $[ML^2T^{-1}]$

41. The dimensions of angular momentum are:

(a) $[ML^2T^{-1}]$ **(b)** $[ML^2T^{-2}]$ **(c)** $[MLT^{-2}]$ **(d)** $[MLT^{-1}]$

42. If radius is doubled at constant speed, centripetal force becomes:

(a) half **(b)** four times **(c)** double **(d)** unchanged

43. If speed is doubled at constant radius, centripetal force becomes:

(a) unchanged **(b)** double **(c)** four times **(d)** half

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44. The frequency of a rotating body is related to angular speed by:

(a) $\omega = 4\pi f$ **(b)** $\omega = f/2\pi$ **(c)** $\omega = \pi f$ **(d)** $\omega = 2\pi f$

45. The period T is related to angular speed by:

(a) $T = \omega/2\pi$ **(b)** $T = \pi\omega$ **(c)** $T = 2\pi\omega$ **(d)** $T = 2\pi/\omega$

46. A satellite close to Earth has period proportional to:

(a) r/GM **(b)** $\sqrt{(r^3/GM)}$ **(c)** r^2/GM **(d)** GM/r

47. The escape velocity from Earth is about:

(a) 11.2 km s^{-1} **(b)** 7.9 km s^{-1} **(c)** 3.0 km s^{-1} **(d)** 1.0 km s^{-1}

48. Artificial gravity frequency for radius R is:

(a) g/R **(b)** $2\pi\sqrt{(g/R)}$ **(c)** $(1/2\pi)\sqrt{(g/R)}$ **(d)** $\sqrt{(gR)}$

49. The motion of a satellite in circular orbit is:

(a) unaccelerated **(b)** accelerated **(c)** uniform velocity **(d)** linear

50. The radian is preferred over degree because it:

(a) simplifies angular equations **(b)** is smaller **(c)** has dimensions **(d)** is larger

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Answer Key with Brief Reasons

1. Answer: (a) 57.3° — 180/π ≈ 57.3°.

2. Answer: (d) 2π rad — Full circle is 2π rad.

3. Answer: (b) radian — Angle in radians.

4. Answer: (c) v = rω — Arc speed = radius × angular speed.

5. Answer: (d) a_t = rα — Tangential acceleration = rα.

6. Answer: (a) v²/r — a_c = v²/r.

7. Answer: (c) mv²/r — F_c = mv²/r.

8. Answer: (c) toward the centre — It is inward radial force.

9. Answer: (c) constant — Only direction changes.

10. Answer: (a) changing — Direction changes continuously.

11. Answer: (b) zero — Force is perpendicular to displacement.

12. Answer: (a) gravity — Earth's gravity supplies inward force.

13. Answer: (c) friction — Static friction acts inward.

14. Answer: (b) √(GM/r) — Gravity = centripetal gives v = √(GM/r).

15. Answer: (c) mass of Earth and radius — Satellite mass cancels.

16. Answer: (c) 7.9 km s⁻¹ — Low orbit speed ≈ 7.9 km/s.

17. Answer: (c) 84 min — Low orbit period ≈ 84 min.

18. Answer: (c) they are in free fall — Satellite and astronaut fall together.

19. Answer: (d) normal force — Wall pushes astronaut inward.

20. Answer: (d) mass distribution about axis — I = Σmr².

21. Answer: (d) moment of inertia — I resists angular acceleration.

22. Answer: (a) rF sin θ — τ = rF sin θ.

23. Answer: (c) τ = Iα — Torque = moment of inertia × angular acceleration.

24. Answer: (a) Iω — L = Iω.

25. Answer: (a) external torque is zero — No external torque means L constant.

26. Answer: (b) increases — I decreases, so ω increases.

27. Answer: (b) decreases — Mass moves closer to axis.

28. Answer: (a) angular momentum — Large L resists change of axis.

29. Answer: (a) rotational kinetic energy — E = ½Iω².

30. Answer: (c) centripetal force — Denser particles need larger inward force.

31. Answer: (a) different at different radii — v = rω depends on r.

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32. Answer: (a) zero — Displacement is zero.

33. Answer: (c) zero — Velocity returns to initial value.

34. Answer: (a) maximum — $T = mv^2/r + mg$ at bottom.

35. Answer: (a) neck — Light bubbles move toward centre of rotation.

36. Answer: (b) infinite — Effective g is zero in free fall.

37. Answer: (d) rad s^{-2} — $\alpha = \Delta\omega/\Delta t$.

38. Answer: (c) N m — Torque = force × distance.

39. Answer: (a) kg m^2 s^{-1} — $L = I\omega$.

40. Answer: (b) $[ML^2]$ — $I = \Sigma mr^2$.

41. Answer: (a) $[ML^2T^{-1}]$ — $L = I\omega$.

42. Answer: (a) half — $F \propto 1/r$.

43. Answer: (c) four times — $F \propto v^2$.

44. Answer: (d) $\omega = 2\pi f$ — One revolution is 2π rad.

45. Answer: (d) $T = 2\pi/\omega$ — $T = 2\pi/\omega$.

46. Answer: (b) $\sqrt{(r^3/GM)}$ — $T = 2\pi\sqrt{(r^3/GM)}$.

47. Answer: (a) 11.2 km s^{-1} — Escape speed ≈ 11.2 km/s.

48. Answer: (c) $(1/2\pi)\sqrt{(g/R)}$ — $f = (1/2\pi)\sqrt{(g/R)}$.

49. Answer: (b) accelerated — Direction changes, so it accelerates.

50. Answer: (a) simplifies angular equations — Calculus and angular relations are natural in radians.

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